

First Term Examination - 2080

Class : XI

Subject: Mathematics (0071)

Full Marks: 75

Time: 3 hrs.

SET - A

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

[11 × 1 = 11]

Rewrite the best alternative of each question in your answer sheet.

1. If $A = \{a, b, c\}$, then the number of non-empty proper subsets of A is
(a) 5 (b) 6 (c) 7 (d) 8
2. If $A \cap B = \phi$, then $A \cap \bar{B} =$
(a) A (b) B (c) $\bar{B}$ (d) $A \cap B$
3. If x, y are real numbers and $|x| + |y| = 0$, then
(a) $x > 0, y < 0$ (b) $x < 0, y > 0$
(c) $x = 0, y = 0$ (d) $x = 0, y \neq 0$
4. Let $A = \{a, b, c\}$. Then $f = \{(a, b), (b, c), (c, a)\}$ is
(a) not a function from A to A (b) one –one but not onto function
(c) onto but not one-one function (d) both one to one and onto function
5. If $A = \begin{bmatrix} 4 & x+3 \\ 2x-1 & -1 \end{bmatrix}$ be a symmetric matrix, then the value of x is
(a) $x = 1$ (b) $x = 2$ (c) $x = 3$ (d) $x = 4$
6. If A and B be two matrices of order 2×2 such that $|A| = -1$ and $|B| = 5$, then $|2AB|$ equals
(a) 10 (b) -10 (c) -20 (d) 20

7. The adjoint of matrix $A = \begin{bmatrix} 2 & -3 \\ 6 & 1 \end{bmatrix}$ is
- (a) $\begin{bmatrix} 1 & -6 \\ 3 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3 \\ -6 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 3 \\ 1 & 6 \end{bmatrix}$ (d) $\begin{bmatrix} 6 & 3 \\ 1 & 2 \end{bmatrix}$
8. If roots of the equation $ax^2 + bx + c = 0$ are negative of each other, then
- (a) $c = 0$ (b) $b = c = 0$ (c) $b = 0$ (d) $a = 0$
9. $\lim_{x \rightarrow 0} \frac{|x|}{x} =$
- (a) 0 (b) 1 (c) -1 (d) does not exist
10. The value of $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$ is
- (a) $\frac{\pi}{180}$ (b) $\frac{180}{\pi}$ (c) 1 (d) 0
11. The point of discontinuity of the function $f(x) = \frac{x^2 - 4x + 4}{x - 2}$ is
- (a) 1 (b) 2 (c) 3 (d) 4

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. Define Union of two sets in set builder form. If A, B and C are subsets of universal set U, prove that:
- i) $A - (B \cup C) = (A - B) \cap (A - C)$
- ii) $(A \cup B) - C = (A - C) \cup (B - C)$. [1+2+2]
13. a) Define the absolute value of the real number. [1]
- b) Prove that $|x + y| \leq |x| + |y|$ for any two real numbers x and y . [2]
- c) Solve the inequality $|2x - 1| \leq 3$ and draw its graph. [2]
14. a) Let $f: A \rightarrow \mathbb{R}$ be given by $f(x) = 2|x| + 2$, where $A = \{-2, 0, 1, 2\}$, find the range of f . [2]
- b) Define skew-symmetric matrix with an example of order 3. If this matrix is A, find $A + A^T$ and comment on the result. [1+2]

15. a) Using the properties of determinant, prove that:

$$\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = (x+y+z)(z-x)^2 \quad [3]$$

- b) For what value of a will the quadratic equation

$$x^2 - (3a-1)x + 2(a^2-1) = 0 \text{ have equal roots.} \quad [2]$$

16. a) Show that the roots of the equation

$$2(a^2 + b^2)x^2 + 2(a+b)x + 1 = 0 \text{ can never be real unless } a = b. [2]$$

- b) Write Euler's constant (e) in terms of a limit as $x \rightarrow 0$. Prove that

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \quad [1+2]$$

17. Write any three indeterminate forms. Evaluate $\lim_{x \rightarrow \theta} \frac{x \cos \theta - \theta \cos x}{x - \theta}$ [1+4]

18. a) Define continuity of a function at a point. [1]

- b) A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} \frac{3x^2-27}{x-3} & \text{for } x \neq 3 \\ p & \text{for } x = 3 \end{cases}$$

Find the value of p if the function is continuous at $x = 3$. [2]

- c) Test the continuity of $f(x) = |x|$ at $x = 0$. [2]

19. a) Let $f(x) = \frac{x - \sqrt{2-x^2}}{2x - \sqrt{2+2x^2}}$.

(i) Find $f(1)$ and comment on the result. [1]

(ii) Evaluate $\lim_{x \rightarrow 1} f(x)$ [2]

- b) Evaluate: $\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a}$ [2]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) When are two statements logically equivalent? Prove that $\sim (p \vee q)$ and $(\sim p) \wedge (\sim q)$ are logically equivalent. Write negation of the statement: 11 is odd or $2 \times 10 = 30$. [1+2+1]
- b) Define lower triangular matrix. Give an example of lower triangular matrix A of size 3×3 . Find the transpose of A and mention the type of A^T with reason. [1+1+1+1]
21. Write the relation between the minor and cofactor of an element of a square matrix. [1]
- If $A = \begin{bmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 4 & -7 \end{bmatrix}$, then
- a) write the elements a_{21}, a_{22}, a_{23} of the matrix A. [1]
- b) find the minors M_{21}, M_{22}, M_{23} of the elements a_{21}, a_{22}, a_{23} . [2]
- c) find the cofactors A_{21}, A_{22}, A_{23} of the elements a_{21}, a_{22}, a_{23} . [1]
- d) find the value of $a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$. [1]
- e) find the value of $|A|$. [1]
- f) show that $a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} = |A|$. [1]
22. Let $g(x) = 2x + 1$ and $h(x) = \sqrt{x}$,
- a) find $(hog)(x)$ and $(goh)(x)$ [1+1]
- b) find the derivative of $f(x) = (hog)(x)$ from first principle. [3]
- c) Evaluate: $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x} - \sqrt{x-a})$ [3]

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